

Mridul Negi

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Computer Engineering | CGPA: 8.01/10 (till Sem VI)

Patiala, Punjab

Aug. 2023 – May 2027

Bhupindra International Public School

Senior Secondary (Class XII), CBSE | 92.8%

Patiala, Punjab

2023

EXPERIENCE

Intern, ELC Summer Program

Thapar Institute of Engineering and Technology

Patiala, Punjab

Jun. – Jul. 2025

- Engineered a prototype to capture real-time traffic data using IoT sensors and cameras.
- Applied SAC-based reinforcement learning for adaptive traffic signal control, improving throughput by 75% and doubling average vehicle speed in rush-hour simulations.
- Reduced average trip time by 30% versus fixed-timer baselines, validated across 4 SUMO traffic scenarios (light, medium, heavy, rush hour), with a web dashboard for live KPI monitoring.

PROJECTS

Mahoraga – Adaptive Combat RL Environment | Python, FastAPI, Docker, LoRA

Apr. 2026 · Finalist

- Built and containerized a FastAPI backend (`reset/step/model-status`) serving a 4-bit-quantized Qwen2.5-3B LoRA agent with a rule-based fallback; deployed to Hugging Face Spaces via a multi-stage Docker build.
- Fine-tuned Qwen2.5-3B via LoRA using an iterative reward-weighted SFT loop (Unsloth + TRL) with rejection sampling and expert-trajectory seeding across 5 training iterations.
- Developed a React dashboard and 33 unit tests covering environment mechanics and reward logic; team reached the national in-person finals of a 36-hour hackathon.

PlantBrain – Industrial Knowledge RAG | FastAPI, pgvector, NetworkX, RAG

Jul. 2026 · Backend & AI lead

- Led the backend and applied-AI core (85% of ~3.7K backend LOC, 19 endpoints) of a cited retrieval system that answers plain-English questions about plant assets from manuals, work orders, and incident logs.
- Built a GraphRAG pipeline combining pgvector semantic search with a NetworkX knowledge graph for multi-hop asset-to-failure links, plus three grounded agents for root-cause, compliance-gap, and lessons-learned analysis.
- Enforced no-source-no-answer and prompt-injection guardrails with structured cited JSON output; measured 100% top-5 retrieval hit-rate and 0.82 answer faithfulness (RAGAS), cutting lookup time from ~12 min to ~4 s.

Cosmic Bot – Community Discord Bot | Python, py-cord, PostgreSQL, Lavalink

2021 – Present

- Built and maintained a production Discord bot (py-cord, 6K+ LOC, 25 modular cogs, 70+ commands) serving a live community for over 4 years.
- Engineered a Lavalink/wavelink music pipeline and migrated core database access from synchronous psycopg2 to async connection pooling (aiopg/asyncpg) to prevent event-loop blocking.
- Shipped DB-backed leveling with dynamically Pillow-rendered rank cards, moderation tooling, and a hot cog load/unload/reload admin interface.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, React, py-cord, WebSockets, SQLAlchemy, Pydantic

Databases: PostgreSQL, SQLite

Developer Tools: Git, Docker, Google Cloud Platform (GCP), Hugging Face Spaces, Render

AI & ML: LLM fine-tuning (LoRA, Unsloth, TRL), LLM agents (Google ADK, Gemini), RL, Computer Vision

CERTIFICATIONS

Fundamentals of Accelerated Computing with CUDA Python | NVIDIA

Apr. 2026

Getting Started with AI on Jetson Nano | NVIDIA

Feb. 2026

Getting Started with Accelerated Computing in Modern CUDA C | NVIDIA

Feb. 2026

Generative AI with Diffusion Models | NVIDIA

Dec. 2025

Getting Started with Deep Learning | NVIDIA

Sep. 2025